

1 Multiplication

In this section, I want to explore how the add primitive can be used to build multiplication primitive. Multiplication is just repeated addition, so multiplication it is a great case study for how to build more meaning using narratives built from primitives. However, clearly I can establish multiplication on it own as a primitive because its essentially a primitive operator and I'm sure there are optimizations, it is just a very simple case study that shows how meaning can be built on top of primitives.

So despite me establishing multiply as a primitive, if I were to proceed with designing multiplication using repeated addition, I would do it as follows:

Multiply operand1 and operand2.

Store 0 in result. Store 0 in counter. While the counter is less than operand2, store result of add operand1 and result in result then store result of 1 and counter in counter.

It is clear from this that to create the design that realizes this narrative, you need to establish how while will be automatically translated into a design structure. In this case, it is simply a conditional check that counter is less than the operand and apply the subsequent store operations. This can be automatically constructed as a design from the narrative and then an engine can reason about it.

Note that I don't specify the invariants in the narrative, since the transformations are built from primitives, the invariants are established through construction.